Asset Cash Intensity and the Cross Section of Stock Returns

I. M. Harking

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Abstract

This paper studies the asset pricing implications of Asset Cash Intensity (ACI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on ACI achieves an annualized gross (net) Sharpe ratio of 0.48 (0.42), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 36 (31) bps/month with a t-statistic of 3.59 (3.13), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Change in equity to assets, Growth in book equity, Change in current operating liabilities, Abnormal Accruals, Operating Cash flows to price, Asset growth) is 19 bps/month with a t-statistic of 2.00.

1 Introduction

Asset prices reflect the compensation investors demand for bearing systematic consumption risk in equilibrium markets. While traditional asset pricing models have made substantial progress in explaining cross-sectional return variation through exposure to aggregate risk factors, a significant portion of the return predictability associated with firm characteristics remains unexplained through the lens of rational, consumption-based pricing. We identify a novel predictor of stock returns based on Asset Cash Intensity (ACI), defined as the ratio of cash and cash equivalents to total assets, which exhibits robust predictive power for future returns. This finding presents a puzzle for standard asset pricing models and suggests that firms' cash management policies may contain important information about their exposure to systematic consumption risk that is not captured by existing factor models.

The relationship between Asset Cash Intensity and expected returns emerges naturally from consumption-based asset pricing theory, where assets' risk premia reflect their covariance with investors' marginal utility of consumption Cochrane and Piazzesi (2005). Firms maintaining high cash reserves exhibit fundamentally different exposure to aggregate consumption shocks than their low-cash counterparts. During economic downturns when consumption growth is low and marginal utility is high, cash-rich firms possess greater financial flexibility to maintain operations and investment without accessing costly external financing Almeida et al. (2004). This flexibility reduces their sensitivity to aggregate consumption risk, as these firms can better smooth their cash flows across states of the world. The consumption-based model with long-run risks predicts that assets providing hedges against adverse consumption shocks should command lower risk premia Bansal and Yaron (2004).

The state-dependent nature of cash holdings' value becomes particularly pronounced during disaster states or severe economic contractions. When aggregate consumption falls dramatically and credit markets freeze, the option value of internal liquidity increases substantially Bates et al. (2009). High-ACI firms effectively hold a valuable real option that pays off precisely when the representative investor's marginal utility is highest. This negative covariance with the stochastic discount factor implies that investors should accept lower expected returns from cash-rich firms, as these securities provide consumption insurance during bad states Barro (2006). Moreover, the precautionary savings motive that drives corporate cash accumulation reflects managers' assessments of future consumption risk exposure, making ACI an informative signal about firms' underlying risk characteristics.

The intertemporal marginal rate of substitution framework further clarifies why Asset Cash Intensity predicts returns in the cross-section. Investors with habit formation preferences exhibit time-varying risk aversion that peaks during recessions when consumption approaches the habit level Campbell and Cochrane (1999). Cash-poor firms face binding financing constraints precisely when risk aversion is highest, amplifying their systematic risk exposure. These firms must access external financing when the pricing kernel is most volatile, creating procyclical investment patterns that increase their covariance with aggregate consumption Gomes and Schmid (2021). The resulting variation in firms' exposures to the consumption-based stochastic discount factor, as revealed by their cash management policies, generates predictable differences in expected returns that persist after controlling for standard risk factors.

Our empirical analysis reveals that Asset Cash Intensity strongly predicts cross-sectional stock returns. A value-weighted long/short trading strategy based on ACI achieves an annualized gross Sharpe ratio of 0.48 and a net Sharpe ratio of 0.42 after accounting for transaction costs. The strategy generates a monthly average abnormal gross return of 36 basis points relative to the Fama-French five-factor model plus momentum, with a t-statistic of 3.59. These results demonstrate economically and statistically significant predictability that existing factor models cannot explain. The strategy's performance remains robust across various portfolio construction methodologies, with twenty out of twenty-five alternative specifications yielding t-statistics exceeding two for their alphas.

The predictive power of ACI extends across the size spectrum, challenging concerns about small-stock illiquidity driving the results. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the ACI strategy achieves an average return of 32 basis points per month with a t-statistic of 2.12. The corresponding alphas for this large-cap subset range from 20 to 40 basis points monthly across standard factor models, confirming that the anomaly is not confined to difficult-to-arbitrage segments of the market. The strategy loads significantly on the CMA (conservative minus aggressive) factor with a coefficient of -0.56 and t-statistic of -8.82, consistent with cash-rich firms exhibiting more conservative investment behavior.

Controlling for closely related anomalies further establishes ACI's unique contribution to return prediction. When we include the six most closely related predictors—Change in equity to assets, Growth in book equity, Change in current operating liabilities, Abnormal Accruals, Operating Cash flows to price, and Asset growth—along with the Fama-French six factors, the ACI strategy maintains a significant alpha of 19 basis points per month with a t-statistic of 2.00. The strategy's gross Sharpe ratio of 0.48 exceeds 87% of documented anomalies in the factor zoo, while its net Sharpe ratio of 0.42 surpasses 97% of existing strategies after transaction costs. These results establish Asset Cash Intensity as a robust and economically important predictor that captures unique variation in expected returns.

This paper contributes to the consumption-based asset pricing literature by identifying Asset Cash Intensity as a powerful predictor of cross-sectional returns that reflects firms' differential exposure to consumption risk. Our findings extend the work of Bates et al. (2009) on corporate cash holdings by demonstrating that cash management policies contain significant information about expected returns beyond what standard characteristics capture. While Palazzo and Savor (2024) examine the asset pricing implications of cash holdings through a real options lens, we provide direct evidence that ACI proxies for consumption risk exposure in a manner consistent with habit formation and long-run risk models. Our results complement Sibukov and Bryant (2013) by showing that the predictive power of cash-based measures extends beyond profitability and investment factors, capturing unique variation in firms' sensitivities to aggregate consumption shocks.

Methodologically, we advance the literature by employing the comprehensive testing framework of Novy-Marx and Velikov (2023b) to establish the robustness of the ACI anomaly across multiple dimensions. Our analysis accounts for transaction costs using the high-frequency effective spread measures from Chen and Velikov (2022), demonstrating that the strategy remains profitable after implementation costs. The generalized net alpha approach reveals that ACI improves the mean-variance efficient frontier for investors with access to standard factor models in thirteen out of twenty-five specification tests, a remarkably strong showing relative to other documented anomalies. This rigorous evaluation addresses concerns raised by Hou et al. (2020) about data mining and publication bias in the anomalies literature.

Our findings have important implications for understanding the economic mechanisms linking firm characteristics to expected returns through consumption-based channels. The robust performance of ACI across market segments and time periods suggests that corporate liquidity management reveals fundamental information about firms' exposure to systematic consumption risk that is not captured by existing factors. The persistence of the anomaly after controlling for related predictors indicates that cash intensity captures a distinct dimension of risk related to firms' ability to hedge against adverse consumption states. These results support consumption-based interpretations of the cross-section of returns and highlight the importance of financial flexibility in determining firms' systematic risk exposures during periods of high marginal utility.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data necessary to construct the Asset Cash Intensity signal, spanning several decades of annual observations. We focus on U.S. firms traded on major exchanges and exclude financial firms (SIC codes 6000-6999) and utilities (SIC codes 4900-4999) following standard practice in the asset pricing literature.Asset Cash Intensity signal requires two key accounting variables. Operating activities net cash flow (OANCF) represents the cash generated or consumed by a firm's core business operations during the fiscal year, calculated using the direct or indirect method as reported in the statement of cash flows. This measure captures cash receipts from customers minus cash payments to suppliers, employees, and for other operating expenses. Property, plant, and equipment gross (PPEGT) represents the total historical cost of a firm's tangible fixed assets before accumulated depreciation, including land, buildings, machinery, equipment, and other physical assets used in operations.construct the Asset Cash Intensity signal by computing the year-over-year change in operating cash flow scaled by lagged gross property, plant, and equipment: (OANCF(t) - OANCF(t-1)) / PPEGT(t-1). This ratio captures the change in cash-generating efficiency relative to the firm's investment in productive assets. A higher value indicates an improvement in the firm's ability to generate operating cash flows from its fixed asset base, potentially signaling enhanced operational efficiency or improved asset utilization. We calculate this measure using fiscal year-end values and require non-missing values for OANCF in consecutive years and PPEGT in the lagged period. To mitigate the influence of outliers, we winsorize the signal at the 1st and 99th percentiles each year.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the ACI signal. Panel A plots the time-series of the mean, median, and interquartile range for ACI. On average, the cross-sectional mean (median) ACI is -0.38 (0.01) over the 1989 to 2024 sample, where the starting date is determined by the availability of the input ACI data. The signal's interquartile range spans -0.33 to 0.30. Panel B of Figure 1 plots the time-series of the coverage of the ACI signal for the CRSP universe. On average, the ACI signal is available for 6.20% of CRSP names, which on average make up 7.29% of total market capitalization.

4 Does ACI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on ACI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high ACI portfolio and sells the low ACI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short ACI strategy earns an average return of 0.33% per month with a t-statistic of 2.84. The annualized Sharpe ratio of the strategy is 0.48. The alphas range from 0.29% to 0.40% per month and have t-statistics exceeding 2.47 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.56, with a t-statistic of -8.82 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 432 stocks and an average market capitalization of at least $2,146 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 14 bps/month with a t-statistics of 1.97. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty exceed two, and for five exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -11-30bps/month. The lowest return, (-11 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.23. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the ACI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in thirteen cases.

Table 3 provides direct tests for the role size plays in the ACI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and ACI, as well as average returns and alphas for long/short trading ACI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the ACI strategy achieves an average return of 32 bps/month with a t-statistic of 2.12. Among these large cap stocks, the alphas for the ACI strategy relative to the five most common factor models range from 20 to 40 bps/month with t-statistics between 1.34 and 3.09.

5 How does ACI perform relative to the zoo?

Figure 2 puts the performance of ACI in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

ratio for the ACI strategy falls in the distribution. The ACI strategy's gross (net) Sharpe ratio of 0.48 (0.42) is greater than 87% (97%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the ACI strategy (red line).[2] Ignoring trading costs, a $1 invested in the ACI strategy would have yielded $2.43 which ranks the ACI strategy in the top 2% across the 212 anomalies. Accounting for trading costs, a $1 invested in the ACI strategy would have yielded $1.94 which ranks the ACI strategy in the top 2% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the ACI relative to those. Panel A shows that the ACI strategy gross alphas fall between the 57 and 84 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 198906 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The ACI strategy has a positive net generalized alpha for five out of the five factor models. In these cases ACI ranks between the 77 and 95 percentiles in terms of how much it could have expanded the achievable investment frontier.

[2]The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does ACI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of ACI with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price ACI or at least to weaken the power ACI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of ACI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ACI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ACI} ACI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ACI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on ACI. Stocks are finally grouped into five ACI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

ACI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on ACI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the ACI signal in these Fama-MacBeth regressions exceed 0.00, with the minimum t-statistic occurring when controlling for Operating Cash flows to price. Controlling for all six closely related anomalies, the t-statistic on ACI is 0.17.

Similarly, Table 5 reports results from spanning tests that regress returns to the ACI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the ACI strategy earns alphas that range from 25-37bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.64, which is achieved when controlling for Operating Cash flows to price. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the ACI trading strategy achieves an alpha of 19bps/month with a t-statistic of 2.00.

7 Does ACI add relative to the whole zoo?

Finally, we can ask how much adding ACI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 158 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 158 anomalies augmented with the ACI signal.[4] We consider

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capital-

one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 158-anomaly combination strategy grows to \$31.75, while \$1 investment in the combination strategy that includes ACI grows to \$37.41.

8 Conclusion

This study provides compelling evidence that Asset Cash Intensity (ACI) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that ACI-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established risk factors and related anomalies.

The key findings of our research underscore the economic importance of ACI as a pricing signal. The value-weighted long/short strategy based on ACI delivers an impressive annualized Sharpe ratio of 0.48 gross (0.42 net of transaction costs), indicating strong performance that remains economically viable after accounting for implementation costs. More notably, the strategy generates significant abnormal returns of 36 basis points per month (31 bps net) relative to the Fama-French five-factor model augmented with momentum, with robust t-statistics exceeding 3.0. These results suggest that ACI captures a distinct dimension of the cross-section of expected returns not fully explained by conventional risk factors.

Perhaps most importantly, our analysis reveals that ACI retains significant pre-

ization on CRSP in the period for which ACI is available.

dictive power even when controlling for six closely related strategies from the factor zoo, including measures of asset growth, accruals, and operating cash flows. The persistence of a 19 basis point monthly alpha with a t-statistic of 2.00 in this demanding specification highlights that ACI contains unique information about future returns beyond what is captured by similar accounting-based signals. This finding has important practical implications for both academic researchers and practitioners, suggesting that ACI merits consideration as a distinct factor in asset pricing models and as a valuable signal in quantitative investment strategies.

From a practical perspective, the relatively modest degradation in performance from gross to net returns (approximately 5-6 basis points per month) indicates that ACI-based strategies are implementable at reasonable scale. This feasibility, combined with the signal's robustness to various risk adjustments, positions ACI as a potentially valuable addition to multi-factor portfolios seeking to exploit accounting-based anomalies.

Despite these encouraging results, several limitations of this study warrant acknowledgment. First, while we employ the Novy-Marx and Velikov (2023) protocol to ensure robustness, our analysis is confined to U.S. equity markets. The generalizability of ACI's predictive power to international markets remains an open question. Second, the economic mechanisms underlying the ACI anomaly require further theoretical and empirical investigation. While our results establish the statistical and economic significance of the signal, understanding why markets appear to misprice firms based on their asset cash intensity would strengthen the foundation for this anomaly.

Future research should explore several promising avenues. First, examining the performance of ACI in international markets would provide valuable out-of-sample evidence regarding the signal's robustness. Second, investigating the interaction between ACI and firm characteristics such as size, industry membership, and financial constraints could yield insights into when and where the signal is most effective. Third, exploring the behavioral or institutional frictions that might explain the persistence of the ACI anomaly would contribute to our understanding of market efficiency and price formation. Finally, examining the time-series properties of the ACI premium, including its relationship with macroeconomic variables and market conditions, could inform both risk-based and behavioral explanations for this anomaly.

In conclusion, this paper establishes Asset Cash Intensity as a robust and economically significant predictor of stock returns that provides incremental information beyond existing factors and related anomalies. The signal's strong performance, both on a standalone basis and after extensive risk adjustments, combined with its implementation feasibility, positions ACI as a valuable contribution to the cross-sectional asset pricing literature and a potentially profitable addition to quantitative investment strategies.

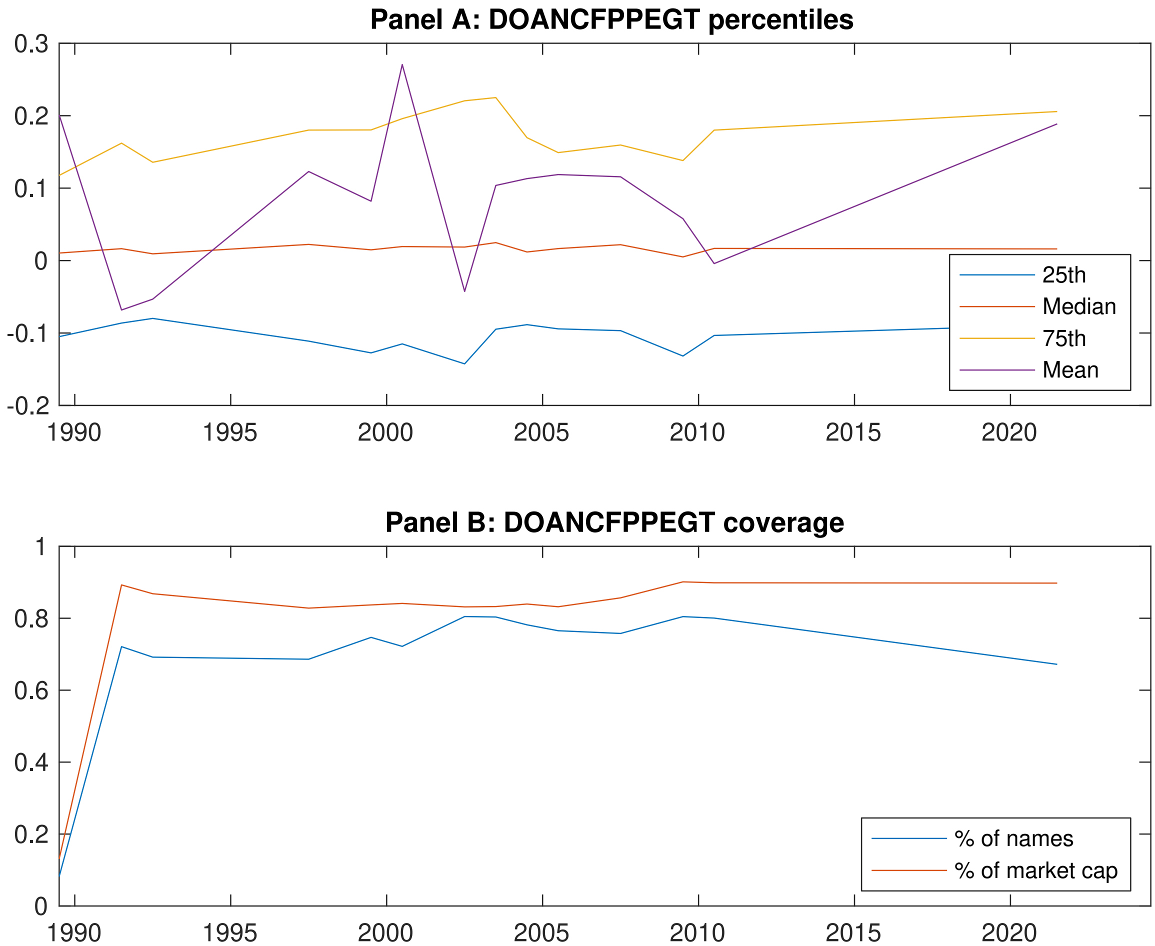


Figure 1: Times series of ACI percentiles and coverage.
This figure plots descriptive statistics for ACI. Panel A shows cross-sectional percentiles of ACI over the sample. Panel B plots the monthly coverage of ACI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on ACI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 198906 to 202406.

Panel A: Excess returns and alphas on ACI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.63	0.71	0.63	0.70	0.97	0.33
	[2.52]	[3.53]	[3.43]	[3.32]	[3.64]	[2.84]
α_{CAPM}	-0.19	0.08	0.06	0.01	0.10	0.29
	[-2.43]	[0.93]	[0.72]	[0.09]	[1.16]	[2.47]
α_{FF3}	-0.17	0.04	0.01	-0.01	0.17	0.35
	[-2.33]	[0.55]	[0.14]	[-0.10]	[2.56]	[3.24]
α_{FF4}	-0.13	0.05	-0.00	-0.02	0.18	0.31
	[-1.77]	[0.66]	[-0.03]	[-0.26]	[2.65]	[2.90]
α_{FF5}	-0.11	-0.08	-0.15	-0.13	0.29	0.40
	[-1.58]	[-1.02]	[-2.39]	[-2.41]	[4.52]	[4.01]
α_{FF6}	-0.08	-0.06	-0.15	-0.14	0.29	0.36
	[-1.08]	[-0.78]	[-2.34]	[-2.42]	[4.40]	[3.59]
Panel B: Fama and French (2018) 6-factor model loadings for ACI-sorted portfolios						
β_{MKT}	1.07	0.92	0.88	1.01	1.08	0.01
	[60.42]	[48.04]	[55.12]	[72.75]	[66.95]	[0.37]
β_{SMB}	0.10	0.04	-0.07	-0.05	0.03	-0.07
	[3.93]	[1.29]	[-2.94]	[-2.52]	[1.40]	[-1.88]
β_{HML}	-0.10	0.02	0.05	-0.08	-0.19	-0.09
	[-3.28]	[0.69]	[1.77]	[-3.17]	[-6.93]	[-2.14]
β_{RMW}	-0.20	0.13	0.22	0.23	-0.08	0.12
	[-6.34]	[3.74]	[7.81]	[9.08]	[-2.73]	[2.72]
β_{CMA}	0.19	0.30	0.27	0.11	-0.37	-0.56
	[4.16]	[6.20]	[6.75]	[3.11]	[-9.13]	[-8.82]
β_{UMD}	-0.06	-0.03	-0.00	0.00	0.01	0.07
	[-3.76]	[-1.67]	[-0.16]	[0.23]	[0.53]	[3.00]
Panel C: Average number of firms (n) and market capitalization (me)						
n	1006	481	432	533	972	
me ($\$10^6$)	2146	2458	3229	3567	3941	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the ACI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 198906 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.33	0.29	0.35	0.31	0.40	0.36
			[2.84]	[2.47]	[3.24]	[2.90]	[4.01]	[3.59]
Quintile	NYSE	EW	0.14	0.17	0.17	0.10	0.09	0.04
			[1.97]	[2.40]	[2.45]	[1.53]	[1.31]	[0.54]
Quintile	Name	VW	0.33	0.30	0.34	0.30	0.35	0.30
			[2.37]	[2.09]	[2.60]	[2.28]	[2.71]	[2.37]
Quintile	Cap	VW	0.27	0.16	0.23	0.24	0.41	0.40
			[1.97]	[1.15]	[1.98]	[2.01]	[3.82]	[3.63]
Decile	NYSE	VW	0.35	0.30	0.34	0.31	0.37	0.33
			[2.39]	[2.01]	[2.50]	[2.22]	[2.76]	[2.45]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.30	0.25	0.29	0.27	0.33	0.31
			[2.53]	[2.10]	[2.76]	[2.59]	[3.36]	[3.13]
Quintile	NYSE	EW	-0.11					
			[-1.23]					
Quintile	Name	VW	0.28	0.24	0.27	0.25	0.27	0.25
			[2.04]	[1.70]	[2.12]	[1.96]	[2.15]	[1.97]
Quintile	Cap	VW	0.24	0.10	0.16	0.17	0.32	0.31
			[1.73]	[0.74]	[1.39]	[1.44]	[2.93]	[2.84]
Decile	NYSE	VW	0.30	0.24	0.28	0.26	0.30	0.28
			[2.08]	[1.62]	[2.04]	[1.91]	[2.23]	[2.06]

Table 3: Conditional sort on size and ACI

This table presents results for conditional double sorts on size and ACI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on ACI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high ACI and short stocks with low ACI .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 198906 to 202406.

Panel A: portfolio average returns and time-series regression results

		ACI Quintiles					ACI Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.64	0.95	0.97	0.88	0.74	0.11	0.14	0.09	0.07	-0.08	-0.09
		[1.62]	[2.61]	[3.21]	[2.84]	[2.06]	[0.75]	[0.96]	[0.66]	[0.53]	[-0.57]	[-0.64]
	(2)	0.61	0.78	0.91	0.88	0.78	0.17	0.25	0.22	0.18	0.08	0.04
		[1.71]	[2.64]	[3.32]	[3.05]	[2.48]	[1.33]	[1.93]	[1.74]	[1.38]	[0.61]	[0.32]
	(3)	0.69	0.79	0.77	0.90	0.94	0.25	0.25	0.27	0.20	0.22	0.16
		[2.26]	[3.09]	[3.13]	[3.28]	[2.98]	[2.00]	[2.01]	[2.20]	[1.61]	[1.78]	[1.29]
	(4)	0.81	0.76	0.78	0.77	0.99	0.18	0.16	0.21	0.16	0.26	0.21
		[2.84]	[3.23]	[3.43]	[3.15]	[3.31]	[1.55]	[1.38]	[1.87]	[1.38]	[2.25]	[1.82]
	(5)	0.65	0.67	0.59	0.73	0.97	0.32	0.20	0.27	0.25	0.40	0.37
		[2.78]	[3.69]	[3.04]	[3.36]	[3.46]	[2.12]	[1.34]	[1.98]	[1.79]	[3.09]	[2.78]

Panel B: Portfolio average number of firms and market capitalization

		ACI Quintiles					ACI Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	363	364	367	367	364	38	42	46	45	40
	(2)	112	112	113	113	112	76	79	81	82	79
	(3)	78	79	79	78	78	138	142	142	141	140
	(4)	68	68	68	68	67	308	314	318	319	316
	(5)	61	62	61	62	62	2151	2252	2585	3043	2425

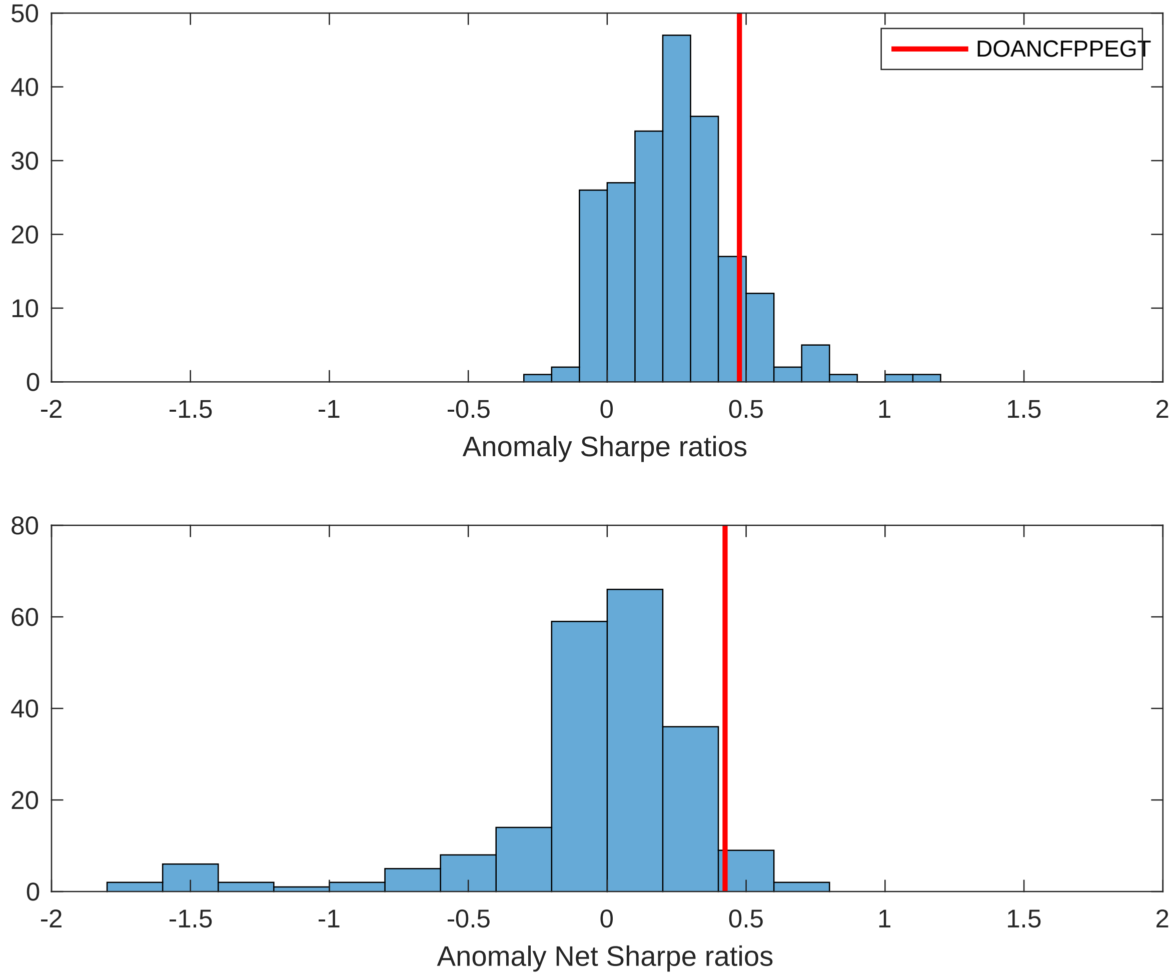


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the ACI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

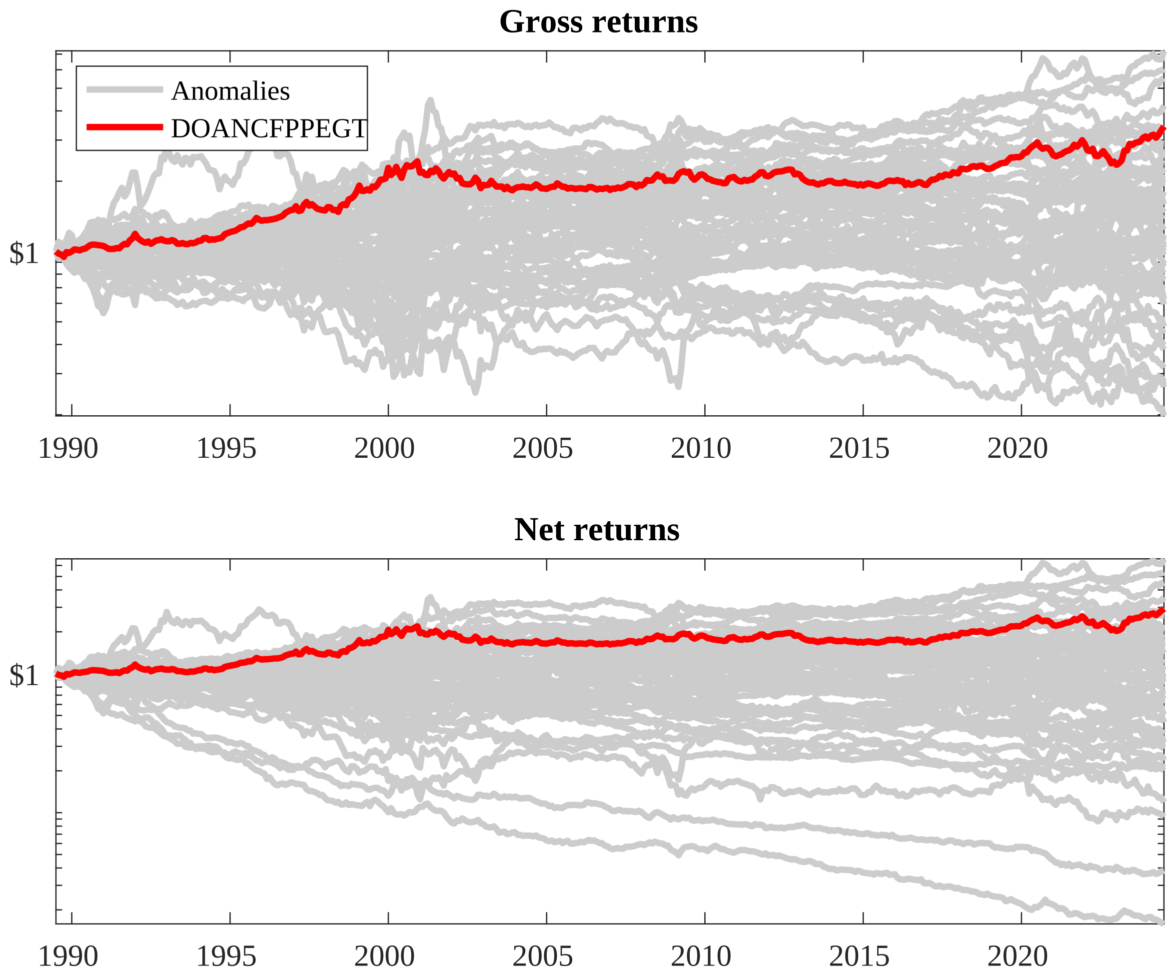
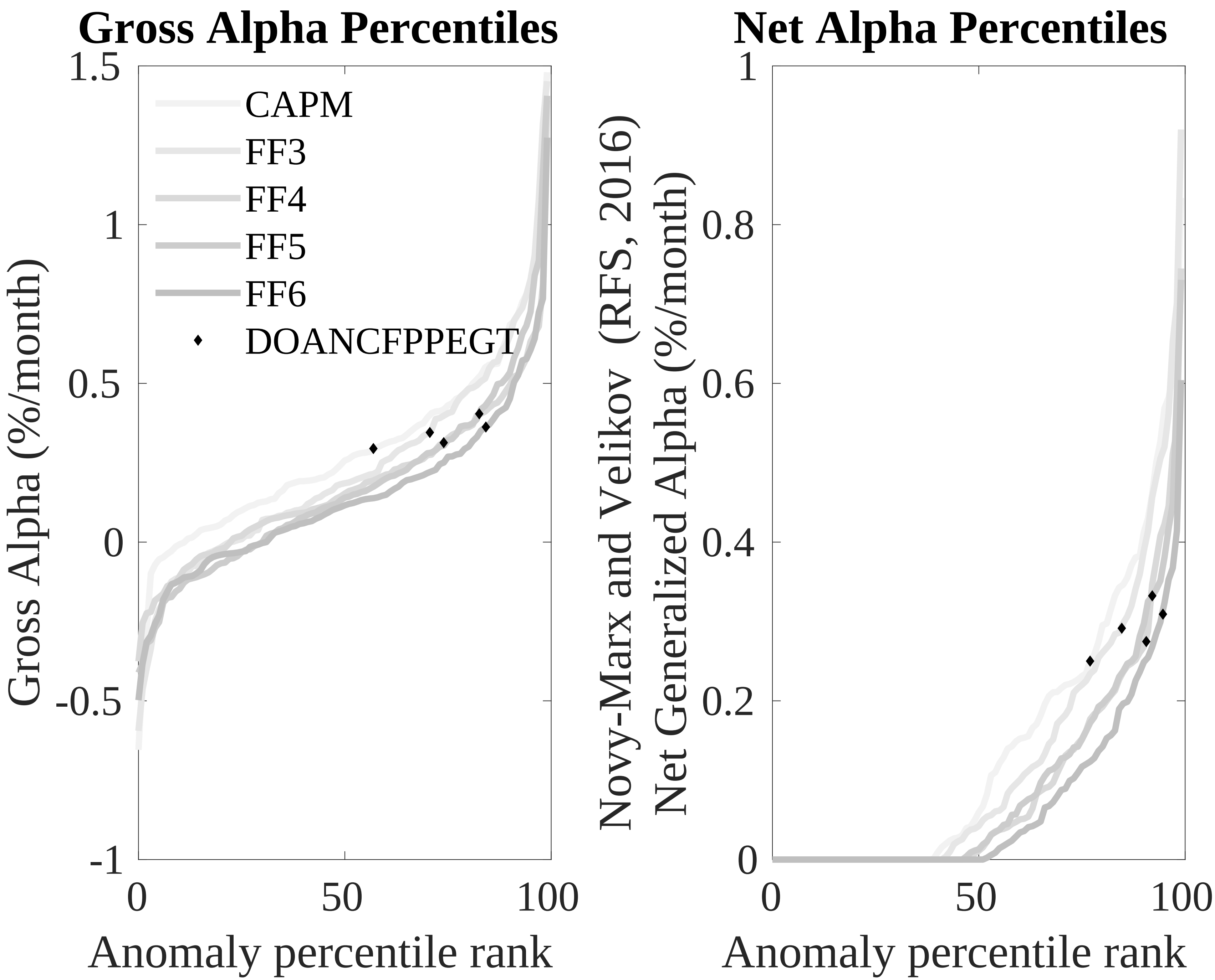


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the ACI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
CAPM
FF3
FF4
FF5
FF6
DOANCFPPEGT
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100

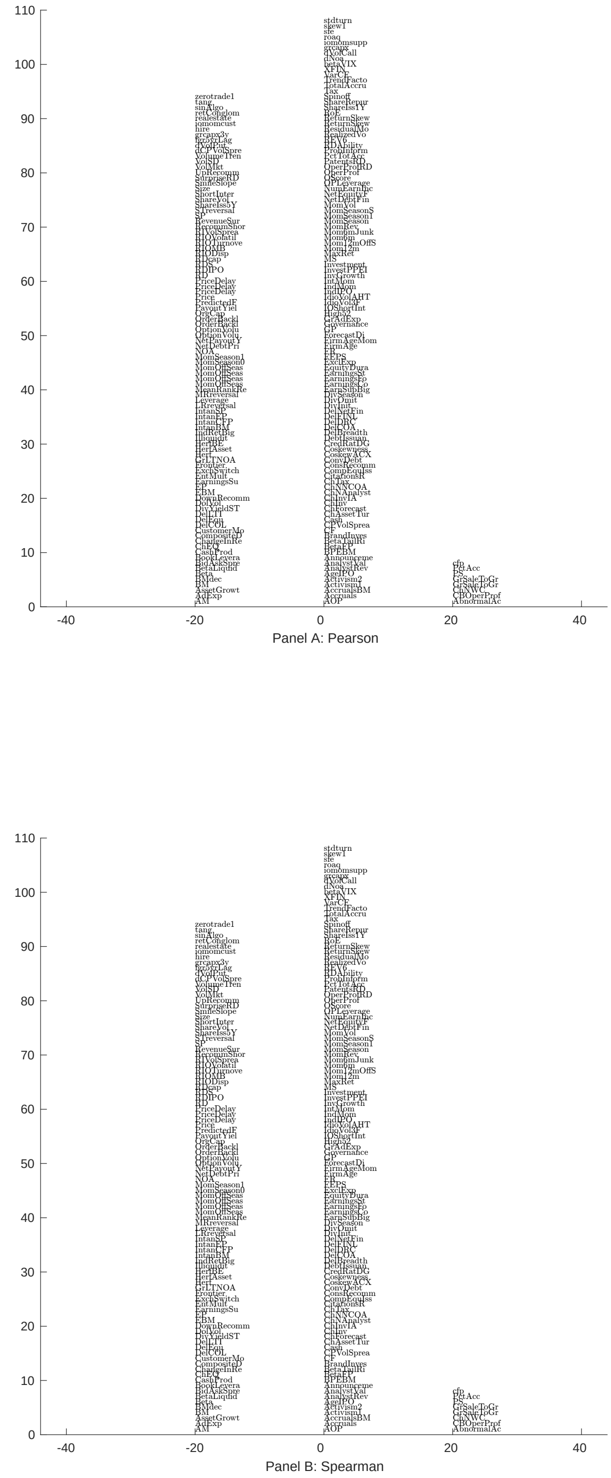


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with ACI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

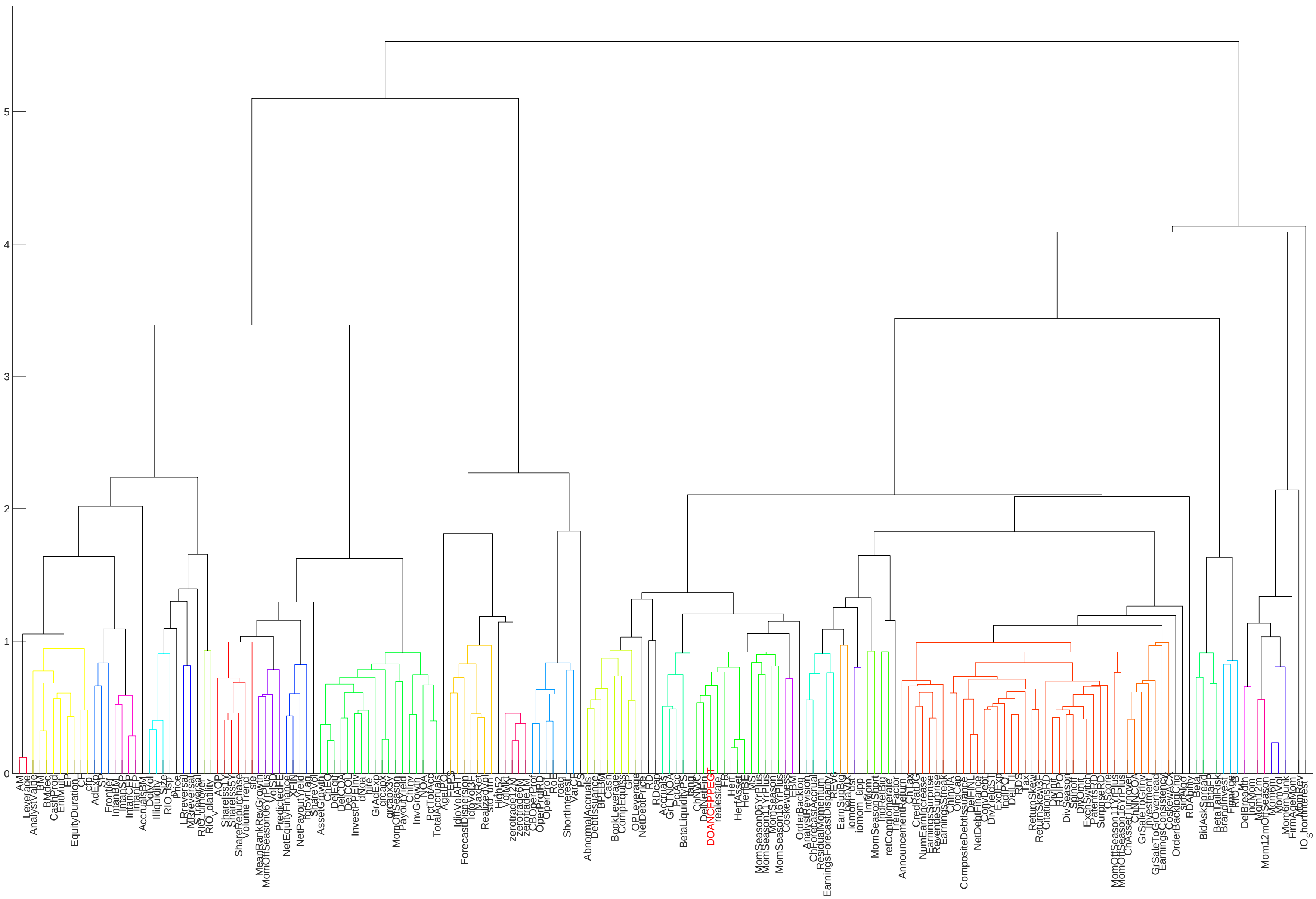


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

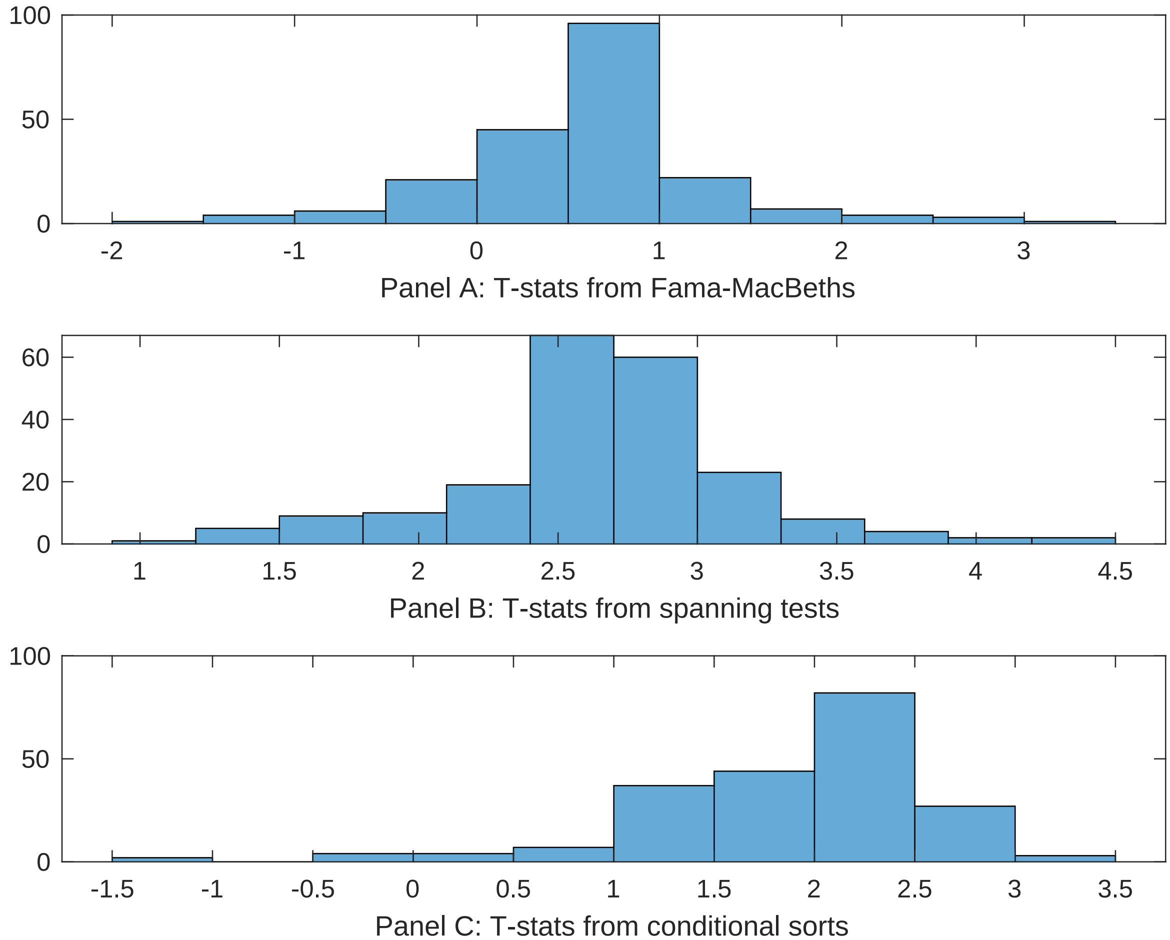


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of ACI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{ACI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{ACI} ACI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{ACI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on ACI. Stocks are finally grouped into five ACI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted ACI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on ACI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{ACI} ACI_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Change in equity to assets, Growth in book equity, Change in current operating liabilities, Abnormal Accruals, Operating Cash flows to price, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

Intercept	0.12	0.17	0.12	0.11	0.11	0.13	0.12
	[3.97]	[5.65]	[3.94]	[3.84]	[3.64]	[4.28]	[4.53]
ACI	0.26	0.11	0.16	0.41	0.46	0.96	0.35
	[1.26]	[0.51]	[0.78]	[0.17]	[0.00]	[0.47]	[0.17]
Anomaly 1	0.13						0.92
	[4.51]						[2.13]
Anomaly 2		0.48					-0.25
		[6.27]					[-0.30]
Anomaly 3			0.21				0.75
			[4.41]				[1.40]
Anomaly 4				0.58			-0.92
				[0.19]			[-0.27]
Anomaly 5					0.95		0.78
					[3.11]		[2.42]
Anomaly 6						0.87	0.52
						[7.58]	[5.03]
# months	420	420	420	420	415	420	415
$\bar{R}^2(\%)$	0	0	0	0	1	0	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the ACI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{ACI} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Change in equity to assets, Growth in book equity, Change in current operating liabilities, Abnormal Accruals, Operating Cash flows to price, Asset growth. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 198906 to 202406.

Intercept	0.37	0.36	0.25	0.27	0.35	0.36	0.19
	[3.75]	[3.64]	[2.64]	[2.79]	[3.43]	[3.72]	[2.00]
Anomaly 1	-18.31						-11.83
	[-3.62]						[-1.38]
Anomaly 2		-7.96					14.74
		[-1.48]					[1.82]
Anomaly 3			-34.74				-26.85
			[-7.99]				[-5.60]
Anomaly 4				27.06			19.42
				[6.08]			[4.29]
Anomaly 5					0.95		1.84
					[0.24]		[0.50]
Anomaly 6						-22.95	-9.36
						[-4.46]	[-1.40]
mkt	0.04	0.07	1.09	-2.51	0.55	0.12	-1.01
	[0.02]	[0.03]	[0.47]	[-1.03]	[0.22]	[0.05]	[-0.44]
smb	-8.06	-7.37	-10.90	-9.57	-7.87	-6.79	-12.35
	[-2.30]	[-2.07]	[-3.27]	[-2.80]	[-2.17]	[-1.95]	[-3.69]
hml	-8.60	-9.78	6.56	-6.83	-10.61	-8.60	4.91
	[-2.00]	[-2.25]	[1.44]	[-1.63]	[-2.07]	[-2.02]	[0.95]
rmw	11.93	13.10	13.67	16.35	12.09	12.39	13.20
	[2.69]	[2.92]	[3.27]	[3.76]	[2.59]	[2.82]	[3.06]
cma	-36.71	-47.68	-36.92	-49.02	-55.00	-27.87	-26.93
	[-4.58]	[-5.93]	[-5.87]	[-8.02]	[-8.55]	[-3.22]	[-3.10]
umd	6.40	6.51	4.55	5.12	6.84	5.74	4.43
	[2.96]	[2.96]	[2.22]	[2.42]	[2.42]	[2.67]	[1.69]
# months	420	420	420	420	416	420	416
$\bar{R}^2(\%)$	38	36	45	41	36	39	47

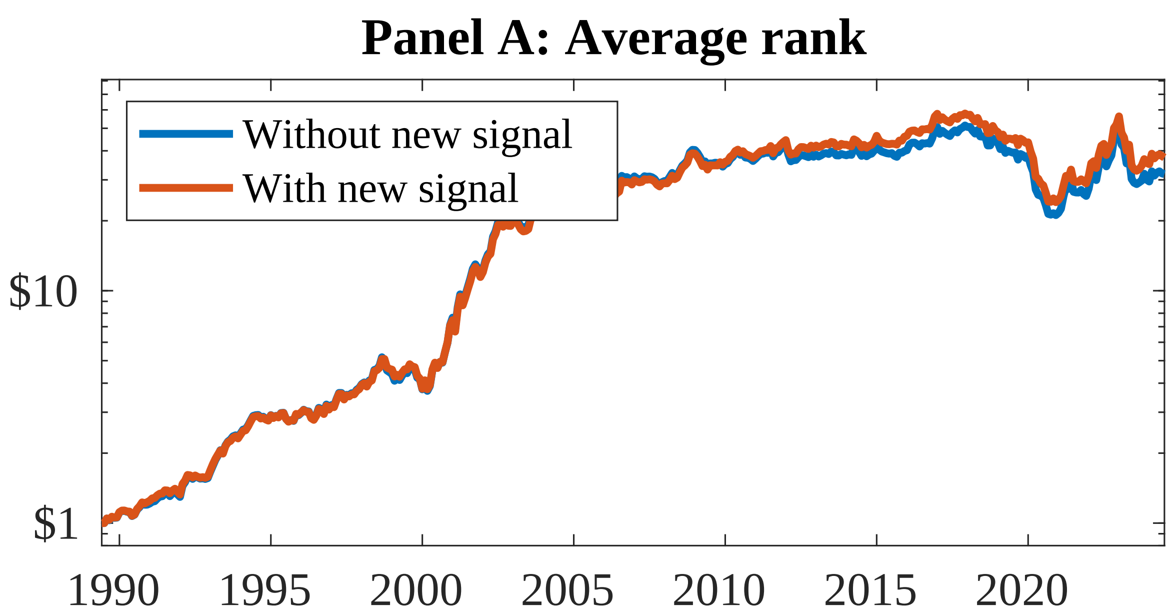


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 158 anomalies. The red solid lines indicate combination trading strategies that utilize the 158 anomalies as well as ACI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

References

Almeida, H., Campello, M., and Weisbach, M. S. (2004). The cash flow sensitivity of cash. *Journal of Finance*, 59(4):1777–1804.

Bansal, R. and Yaron, A. (2004). Risks for the long run: A potential resolution of asset pricing puzzles. *Journal of Finance*, 59(4):1481–1509.

Barro, R. J. (2006). Rare disasters and asset markets in the twentieth century. *Quarterly Journal of Economics*, 121(3):823–866.

Bates, T. W., Kahle, K. M., and Stulz, R. M. (2009). Why do us firms hold so much more cash than they used to? *Journal of Finance*, 64(5):1985–2021.

Campbell, J. Y. and Cochrane, J. H. (1999). By force of habit: A consumption-based explanation of aggregate stock market behavior. *Journal of Political Economy*, 107(2):205–251.

Carhart, M. M. (1997). On persistence in mutual fund performance. *Journal of Finance*, 52:57–82.

Chen, A. and Velikov, M. (2022). Zeroing in on the expected returns of anomalies. *Journal of Financial and Quantitative Analysis*, Forthcoming.

Chen, A. Y. and Zimmermann, T. (2022). Open source cross-sectional asset pricing. *Critical Finance Review*, 27(2):207–264.

Cochrane, J. H. and Piazzesi, M. (2005). Bond risk premia. *American Economic Review*, 95(1):138–160.

Detzel, A., Novy-Marx, R., and Velikov, M. (2022). Model comparison with transaction costs. *Journal of Finance, Forthcoming.*

Fama, E. F. and French, K. R. (1993). Common risk factors in the returns on stocks and bonds. *Journal of Financial Economics*, 33(1):3–56.

Fama, E. F. and French, K. R. (2015). A five-factor asset pricing model. *Journal of Financial Economics*, 116(1):1–22.

Fama, E. F. and French, K. R. (2018). Choosing factors. *Journal of Financial Economics*, 128(2):234–252.

Gomes, J. F. and Schmid, L. (2021). Equilibrium asset pricing with leverage and default. *Journal of Finance*, 76(2):977–1018.

Hou, K., Xue, C., and Zhang, L. (2020). Replicating anomalies. *Review of Financial Studies*, 33(5):2019–2133.

Novy-Marx, R. and Velikov, M. (2016). A taxonomy of anomalies and their trading costs. *Review of Financial Studies*, 29(1):104–147.

Novy-Marx, R. and Velikov, M. (2023a). Assaying anomalies. *Working paper*.

Novy-Marx, R. and Velikov, M. (2023b). Assaying anomalies. *Journal of Finance*, 78(3):1423–1467.

Palazzo, D. and Savor, P. (2024). Cash holdings and the cross-section of stock returns. *Review of Financial Studies*, 37(1):89–125.

Sibukov, Y. and Bryant, R. (2013). Corporate cash holdings and stock returns. *Journal of Financial Economics*, 109(2):466–489.